UNIVERSITY OF TECHNOLOGY, JAMAICA
SCHOOL OF COMPUTING & INFORMATION TECHNOLOGY
Programming 2: Tutorial/Practical Worksheet #1

Tutorial Questions

Part A

Discuss the following questions

1. What is meant by structured programming?
2. Why is it important for instructions in a program to be in the correct order?
3. Name and briefly explain the following control structures:
    a. Sequence
    b. Selection
4. What problems might occur if a program is not structured properly?

Part B

For each scenario

- Identify the condition ***(if any)***
- State the action if the condition is true ***(if any)***
- State the action if the condition is false ***(if any)***
- Indicate whether it represents sequence or selection

1. A user enters a password. If the password is correct, the user is allowed to log in. Otherwise, an error message is displayed.
2. A customer enters the price of an item. The system calculates tax at 15% and displays the total cost.
3. A mobile phone app records the amount of data used each day for five days. The system calculates the total data used, calculates the average daily usage, and displays a summary report.
4. A banking system checks an account balance. If the balance is greater than or equal to the withdrawal amount, the withdrawal is approved. Otherwise, the transaction is declined.
5. A student registration system accepts a student's name, ID number, and three course marks. The system calculates the average mark, assigns the average to a variable, formats a report showing all entered information, and displays the report.

6. A student’s final grade is calculated. If the final score is 70 or higher, the grade is *A*. If the score is between 50 and 69, the grade is *B*. Otherwise, the grade is *Fail*.

Part C

Determine the final value of each variable.

1. x = 5
   x = x + 2
2. x = 10
   y = x % 3
3. x = 4
   y = x + 6
   x = y - 2
4. a = 10
   b = a / 2
   a = b + 5
5. p = 17
   q = p % 5
   p = q + 10
6. x = 3
   y = x + 4
   z = y * 2
   x = z - y
7. a = 20
   b = a / 4
   c = b + a
   a = c - b
8. x = 2
   y = x * 3
   x = y + 4
   y = x - y
9. p = 5
   q = p + 3
   r = q * 2
   p = r – q
   q = p + r
10. m = 29
    n = m % 7
    m = n + 3
    n = m % 4

Part D

Identify the logical error in each question and rewrite it correctly.

1. IF age = 18 THEN
2. IF score > 50 AND score < 40 THEN
3. IF temperature < 20 THEN
      OUTPUT "Hot"
4. IF balance > 0 THEN
      OUTPUT "Account Active"

```
ELSE IF balance > 1000 THEN
    OUTPUT "Premium Account"
ENDIF
```

5.
```
IF score >= 70 THEN
    OUTPUT "A"
ELSE IF score >= 50 THEN
    OUTPUT "C"
ELSE
    OUTPUT "B"
ENDIF
```

6.
```
IF password == "admin" THEN
    OUTPUT "Access Granted"
ENDIF
OUTPUT "Access Denied"
```

In the Laboratory

- Complete any unfinished questions from the tutorial section.
- Log in to the lab computers and familiarize yourself with the system.
- Explore the integrated development environments (IDEs) that will be used in this course for both C and Python

Implement and execute the following programs using C and Python:

1. Write a program that reads a number from input and determines whether it is positive, negative, or zero. Display a message indicating the sign of the number.
2. A website asks a user to enter their age to verify access. If the user is 18 years or older, the program should display “Access Granted.” Otherwise, it should display “Access Denied.” Write a program that reads the age from input and outputs the appropriate message.
3. A banking application must verify a withdrawal request. The account balance must be greater than or equal to the withdrawal amount for the transaction to proceed. If the balance is sufficient, the program should subtract the withdrawal amount from the

balance and display the remaining balance. If the balance is insufficient, it should display an error message.

4. A utility company calculates the electricity bill based on units consumed. The first 100 units cost $0.50 per unit, the next 100 units cost $0.75 per unit, and any units above 200 cost $1 per unit. Write a program that reads the number of units consumed from input, calculates the total bill according to these rules, and displays the total amount.
5. Write a program that reads an integer from input and determines whether the number is even or odd. Additionally, the program should check if the number is divisible by 5. Finally, display the number along with its even/odd status and whether it is divisible by 5
6. A shop wants to calculate the total price of an item including tax. The user enters the price of an item. If the price is less than $100, a tax rate of 5% is applied. Otherwise, a tax rate of 10% is applied. The program should calculate the total price including tax and display it.

If time permits, the scenarios from Part B of the tutorial section may be implemented and executed in both C and Python.